On Reincarnation. Part III: Four Bardo of Consciousness

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Abstract

This paper presents a rigorous logical model describing the states of consciousness of living beings and the continuity of these states through a sequence of lives. According to the proposed model, all living beings always reside in one of three states of consciousness: Wakefulness (W), Dreaming Sleep (DW), or Dreamless Sleep (DS). The state of Bardo of Death (BD) acts as a transitional state leading to a New Bardo of Life (BL'), while Dreamless Sleep normally returns to Wakefulness in the current body with memory preserved. Using the apparatus of predicate logic, Boolean functions, and probabilistic estimates, we formalize transitions between these states and prove the continuity of the stream of consciousness through a sequence of bodies. All considered alternative theories (complete cessation of consciousness after death, existence of other possible states, tautology of Bardo of Death) are refuted as logically inconsistent. The proposed model is supported by empirical observations and is accompanied by a transition graph optimized for analysis using artificial intelligence.

1 Introduction

The phenomenon of consciousness and its presumed continuity, persisting through birth and death, require rigorous and clear logical analysis. This paper develops an axiomatic model of consciousness states, utilizing predicate logic, Boolean functions, and a probabilistic approach, based on research presented in works [1, 2]. According to our model, any living being is in one of three clearly defined states of consciousness: Wakefulness, Dreaming Sleep, or Dreamless Sleep. A special transitional state is the Bardo of Death, fundamentally distinct from Dreamless Sleep in its implicative function, as it initiates the beginning of a new life cycle in a new body. The proposed model does not require metaphysical assumptions and relies entirely on formal logic and observable empirical data, particularly the trend of increasing numbers of living beings. This logical model is optimized for further analysis using artificial intelligence.

Definitions of Bardo States

- **Bardo (general definition)** — a state of consciousness of a conditioned living being involved in the process of becoming (the life–death cycle).

- **Bardo of Life (BL)** — includes Bardo of Wakefulness, Bardo of Dreaming Sleep, and Bardo of Dreamless Sleep:
  - **Bardo of Wakefulness (W)**: Everyday state of consciousness (conscious activity), the waking state in the present life; the interval from conception to death or fatal illness.
  - **Bardo of Dreaming Sleep (DW)**: This state encompasses not only physiological nocturnal sleep accompanied by dreams but also a wide range of altered states of consciousness, including hypnosis, hallucinations, psychoses, and certain forms of meditative trance. Their key characteristic is the presence of internal mental content that does not correspond to current sensory reality. This aligns them with typical Dreaming Sleep, as in both cases, consciousness experiences a subjective reality that may be entirely fictional or partially aligned with reality.
  - **Bardo of Dreamless Sleep (DS, dark sleep)**: A state of deep sleep without dreams, including fainting, anesthesia, coma, and similar states of absent conscious perception. Normally returns to Wakefulness in the current body with memory preserved.

- **Bardo of Death (BD)**: A transitional state of consciousness that inevitably leads to a New Bardo of Life (BL'), opening a new cycle of three states of consciousness (W, DW, DS). Occurs upon the death of a being in any state of the Bardo of Life.

A living being not in one of the four described bardo states logically cannot exist.

2 Definitions and Axioms

2.1 Predicates

For the formal description of consciousness states, bodies, and their properties, the following predicates are used:

- $BL(x)$: Being x is in the Bardo of Life (current cycle of three states: Wakefulness, Dreaming Sleep, Dreamless Sleep).

- $W(x)$: Being x is in the state of Wakefulness in the current body t.

- $DW(x)$: Being x is in the state of Dreaming Sleep in the current body t.

- $DS(x)$: Being x is in the state of Dreamless Sleep in the current body t.

- Exists(x): Being x is a living being involved in the process of becoming.

- Stream(x, t): Being x possesses a stream of consciousness associated with body t.
- Experience(x): Being x has subjective conscious experience (phenomenal consciousness).
- $M(x, t)$: Being x has memory associated with body t.
- $MC(x)$: Being x is in a state of minimal consciousness (e.g., coma).
- $BD(x)$: Being x is in the state of Bardo of Death (transitional state).
- $BL'(x)$: Being x is in the state of New Bardo of Life (beginning of a new cycle in a new body t').

2.2 Sets and Functions

The following sets and functions are introduced:

- **Sets**:
  - $S = \{W(x), DW(x), DS(x)\}$: Set of consciousness states in the current life.
  - $T = \{t_1, t_2, \dots\}$: Set of bodies, where t is the current body, and t' is the new body after transitioning through the Bardo of Death.
- **Boolean Functions (Logical Implications)**:
  - $BL(x) \to BD(x) \equiv \neg BL(x) \lor BD(x)$: The Bardo of Life inevitably leads to the Bardo of Death upon the being's demise.
  - $BD(x) \to BL'(x) \equiv \neg BD(x) \lor BL'(x)$: The Bardo of Death inevitably leads to a New Bardo of Life.
- **Probabilistic Function**:
  - $P(\sigma'|\sigma)$, where $\sigma, \sigma' \in \{W(x), DW(x), DS(x), BD(x), BL'(x)\}$, specifies the probability of transitioning from state σ to state σ'.

2.3 Axioms

Based on works [1, 2], the following axioms are formulated:

1. **Completeness of Bardo of Life States**:

$$\forall x \, (BL(x) \to (W(x) \lor DW(x) \lor DS(x)))$$

   Every living being in the Bardo of Life is in one of three states: Wakefulness, Dreaming Sleep, or Dreamless Sleep.

2. **Exit from Dreamless Sleep**:

$$\forall x \exists t \, (DS(x) \land \text{Stream}(x, t) \rightarrow W(x) \land M(x, t))$$

Dreamless Sleep normally returns the being to Wakefulness in the current body with memory preserved.

3. **Transition Through Bardo of Death**:

$$\forall x \exists t' \, (BD(x) \land \text{Stream}(x, t) \rightarrow BL'(x) \land \text{Stream}(x, t') \land \neg M(x, t))$$

The Bardo of Death leads to a New Bardo of Life in a new body, with the memory of the previous body not preserved.

4. **Inevitability of Bardo of Death**:

$$\forall x \, (BL(x) \rightarrow BD(x))$$

The Bardo of Life, upon the death of the being, transitions to the Bardo of Death.

5. **Transition to New Bardo of Life**:

$$\forall x \, (BD(x) \rightarrow BL'(x))$$

The Bardo of Death always initiates a New Bardo of Life.

6. **Continuity of Stream of Consciousness**:

$$\forall x \exists t_1, t_2 \, (\text{Stream}(x, t_1) \land BD(x) \rightarrow \text{Stream}(x, t_2) \land BL'(x))$$

The stream of consciousness is preserved across the Bardo of Death.

7. **Nonexistence Outside Bardo States**:

$$\forall x \, (\neg \text{Exists}(x) \rightarrow \neg (W(x) \lor DW(x) \lor DS(x) \lor BD(x)))$$

Beings not involved in the process of becoming are not in the described states.

8. **Minimal Consciousness**:

$$\forall x \, ((MC(x) \lor \text{Coma}(x) \lor \text{Hypnosis}(x) \lor \text{LossOfConsciousness}(x)) \rightarrow DS(x))$$

States of minimal consciousness are equivalent to Dreamless Sleep.

9. **Absence of Experiences**:

$$\forall x \, ((DS(x) \lor BD(x)) \rightarrow \neg \text{Experience}(x))$$

In the states of Dreamless Sleep and Bardo of Death, the subject has no subjective experiences.

2.4 Probabilistic Estimates

To formalize transitions, the following probabilities are introduced:

- $P(W(x)|DS(x)) = 1$: Dreamless Sleep normally ends in Wakefulness.
- $P(DW(x)|W(x)) = p_1$, $P(DS(x)|DW(x)) = p_2$: Intralife transitions depend on physiology.
- $P(BD(x)|BL(x)) = 1$: Bardo of Life, upon death, leads to Bardo of Death.
- $P(BD(x)|DS(x)) > 0$: Transition to Bardo of Death is possible upon death in Dreamless Sleep.
- $P(BL'(x)|BD(x)) = 1$: Bardo of Death always leads to a New Bardo of Life.
- $P(\neg\text{Exists}(x)|BD(x)) = 0$: Cessation of existence is excluded.
- $P(D(x)|BD(x)) = 0$: Alternative states are redundant.

2.5 Corollaries

1. **Cyclicity**:
$$BL(x) \to BD(x) \to BL'(x)$$
Consciousness undergoes a cycle of Bardo of Life, Bardo of Death, and New Bardo of Life.

2. **Continuity of Stream**:
$$\forall x \exists t_1, t_2 \, (\text{Stream}(x, t_1) \to \text{Stream}(x, t_2))$$
The stream of consciousness persists through all states.

3. **Distinction Between $DS(x)$ and $BD(x)$**:
   - $DS(x) \to W(x) \wedge M(x, t)$: Dreamless Sleep returns to Wakefulness with memory.
   - $BD(x) \to BL'(x) \wedge \neg M(x, t)$: Bardo of Death leads to a New Bardo of Life without memory.

4. **Death in $DS(x)$**: If a being dies in $DS(x)$, then $DS(x) \to BD(x) \to BL'(x)$.

3 Logical Analysis

3.1 Non-Contradiction

The axiom system is non-contradictory:

- Intralife Transitions:
$$W(x) \leftrightarrow DW(x) \leftrightarrow DS(x)$$
with memory preservation.
- Interlife Transitions:
$$BL(x) \rightarrow BD(x) \rightarrow BL'(x)$$
upon death with memory loss.
- Continuity of Consciousness:
$$BL(x) \rightarrow BD(x) \equiv \neg BL(x) \vee BD(x)$$
$$BD(x) \rightarrow BL'(x) \equiv \neg BD(x) \vee BL'(x)$$
Hence, $BL(x) \rightarrow BL'(x)$.

3.2 Formalization of Memory

Memory is defined as:
$$M(x,t) \equiv \exists m\,(\text{Memory}(x,m) \wedge \text{Link}(m,t))$$
- $DS(x) \rightarrow M(x,t)$: Dreamless Sleep preserves memory.
- $BD(x) \rightarrow \neg M(x,t)$: Bardo of Death erases memory.

3.3 Boundary States

Minimal consciousness is equivalent to Dreamless Sleep:
$$\forall x\,(MC(x) \rightarrow \neg\text{Experience}(x) \rightarrow DS(x))$$
Empirical data confirm the absence of subjective experiences in such states [2].

3.4 Probabilistic Model

A Markov chain is used:
- Set of States: $\Sigma = \{W(x), DW(x), DS(x), BD(x), BL'(x)\}$.
- Probabilities:
  - $P(W(x)|DS(x)) = 1$
  - $P(DW(x)|W(x)) = p_1$
  - $P(DS(x)|DW(x)) = p_2$
  - $P(BD(x)|BL(x)) = 1$
  - $P(BD(x)|DS(x)) > 0$
  - $P(BL'(x)|BD(x)) = 1$

4 Refutation of Counterarguments

1. **Complete Cessation ($BD(x) \rightarrow \neg$Exists(x))**: Excluded, as it contradicts the continuity of consciousness and the observed increase in living beings.

2. **Alternative States ($D(x)$)**: Equivalent to $DS(x)$, $BD(x)$, or $BL'(x)$, or redundant.

3. **Tautology of Bardo of Death ($BD(x) \rightarrow BD(x)$)**: Meaningless, as:
   - It does not explain the origin of life.
   - It contradicts the empirical fact of life's reproduction.
   - $BD(x)$ lacks internal dynamics, making $BD(x) \rightarrow BD(x)$ impossible.
   - Per $BL(x) \rightarrow BD(x) \rightarrow BL'(x)$, Bardo of Death leads to a New Bardo of Life.

5 Transition Graph

- **Nodes**: $\Sigma = \{W(x), DW(x), DS(x), BL(x), BD(x), BL'(x)\}$.
- **Edges**:
  - Intralife:
    - $(W(x), DW(x))$
    - $(DW(x), W(x))$
    - $(DW(x), DS(x))$
    - $(DS(x), DW(x))$
    - $(DS(x), W(x))$
  - Interlife:
    - $(BL(x), W(x))$
    - $(BL(x), DW(x))$
    - $(BL(x), DS(x))$
    - $(BL(x), BD(x))$
    - $(BD(x), BL'(x))$
  - Note: If a being dies in $DS(x)$, the transition $DS(x) \rightarrow BD(x)$ is possible.

6 Conclusion

The presented model is logically consistent, non-contradictory, and supported by empirical observations. The states of consciousness, the distinction between Dreamless Sleep and Bardo of Death, and the process of consciousness transition through life cycles are clearly formalized. This model is optimal for analysis using artificial intelligence.

References

[1] Preece, A., Batenin, B. (Berlin, 2025). On Reincarnation. Part I: Madhyamaka-Prasangika. https://zenodo.org/records/15492409

[2] Preece, A., Batenin, B. (Berlin, 2025). On Reincarnation. Part II: Axiomatic of Implicative Reincarnation. https://zenodo.org/records/15513221

Namaste!